Mini Project: Virtual Windows Desktop on AWS Using Windows Server

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1. Introduction

This mini project involves setting up a virtual Windows desktop using a Windows Server EC2 instance on Amazon Web Services (AWS). It demonstrates the ability to deploy and configure a cloud-based virtual desktop environment, an essential skill for developers and aspiring DevOps engineers.

2. Objective

To gain hands-on experience with cloud infrastructure by deploying a Windows Server instance on AWS, configuring it for remote desktop access, and creating a fully functional virtual desktop environment.

3. Tools & Technologies Used

- Amazon Web Services (AWS)
- EC2 (Elastic Compute Cloud)
- Windows Server
- Remote Desktop Protocol (RDP)

4. What I Did

- Deployed a Windows Server EC2 instance on AWS
- Configured Remote Desktop access for secure connectivity - Set up a functional virtual desktop environment in the cloud

- 4.1

- “All credentials are go. Preparing to download the RDP file and connect to the AWS EC2 instance i-06be86f961563dae3."

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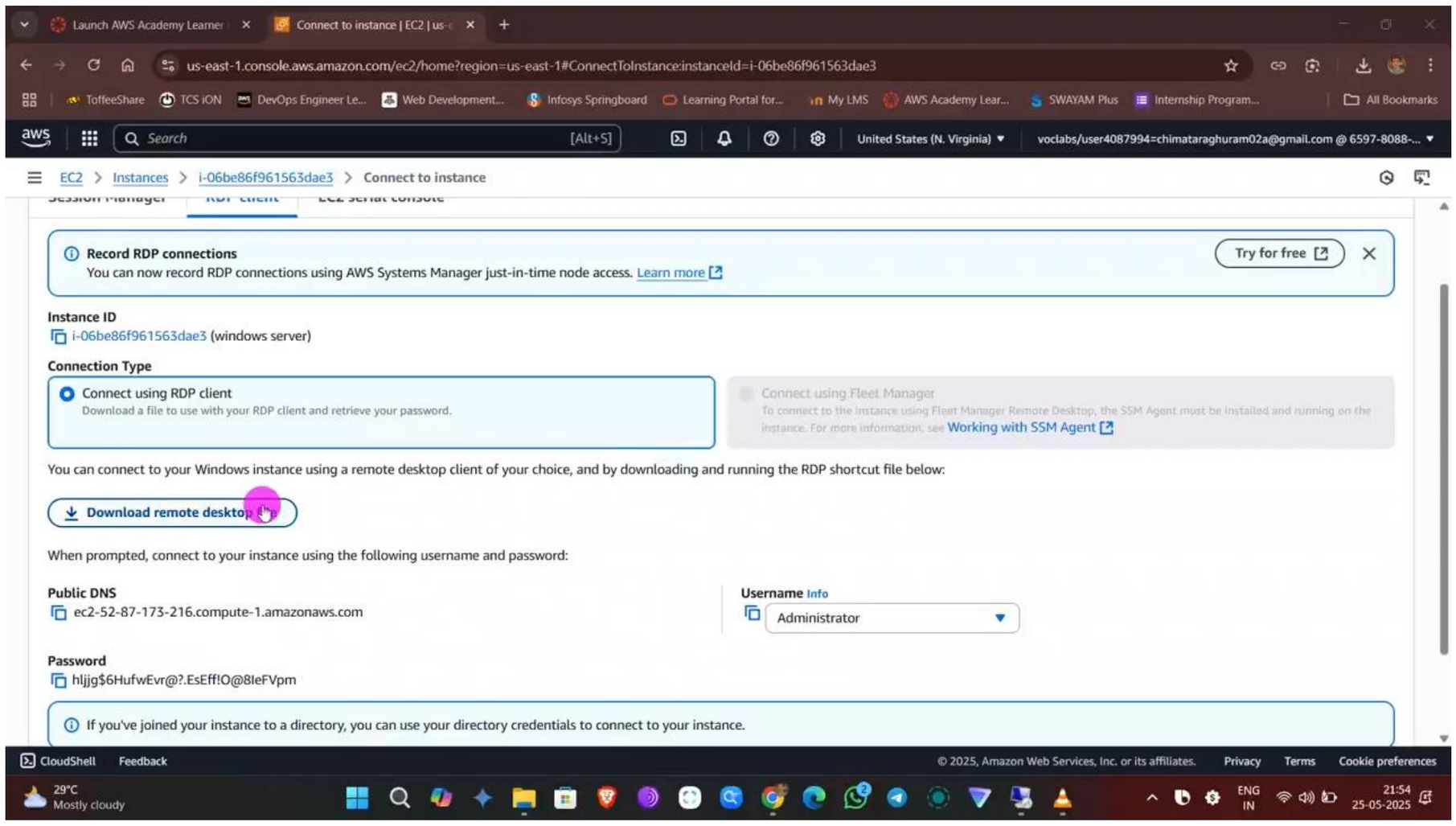
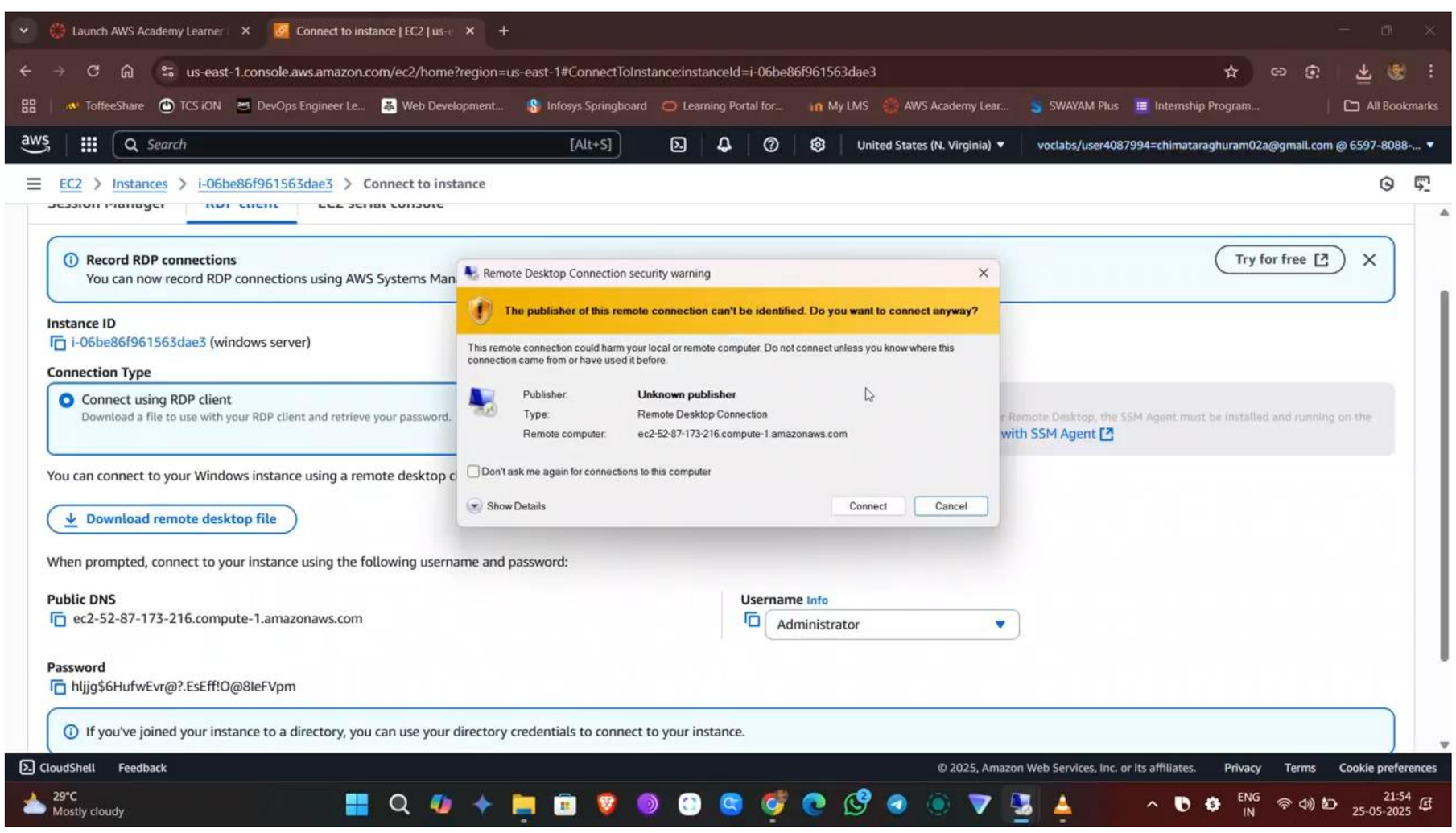


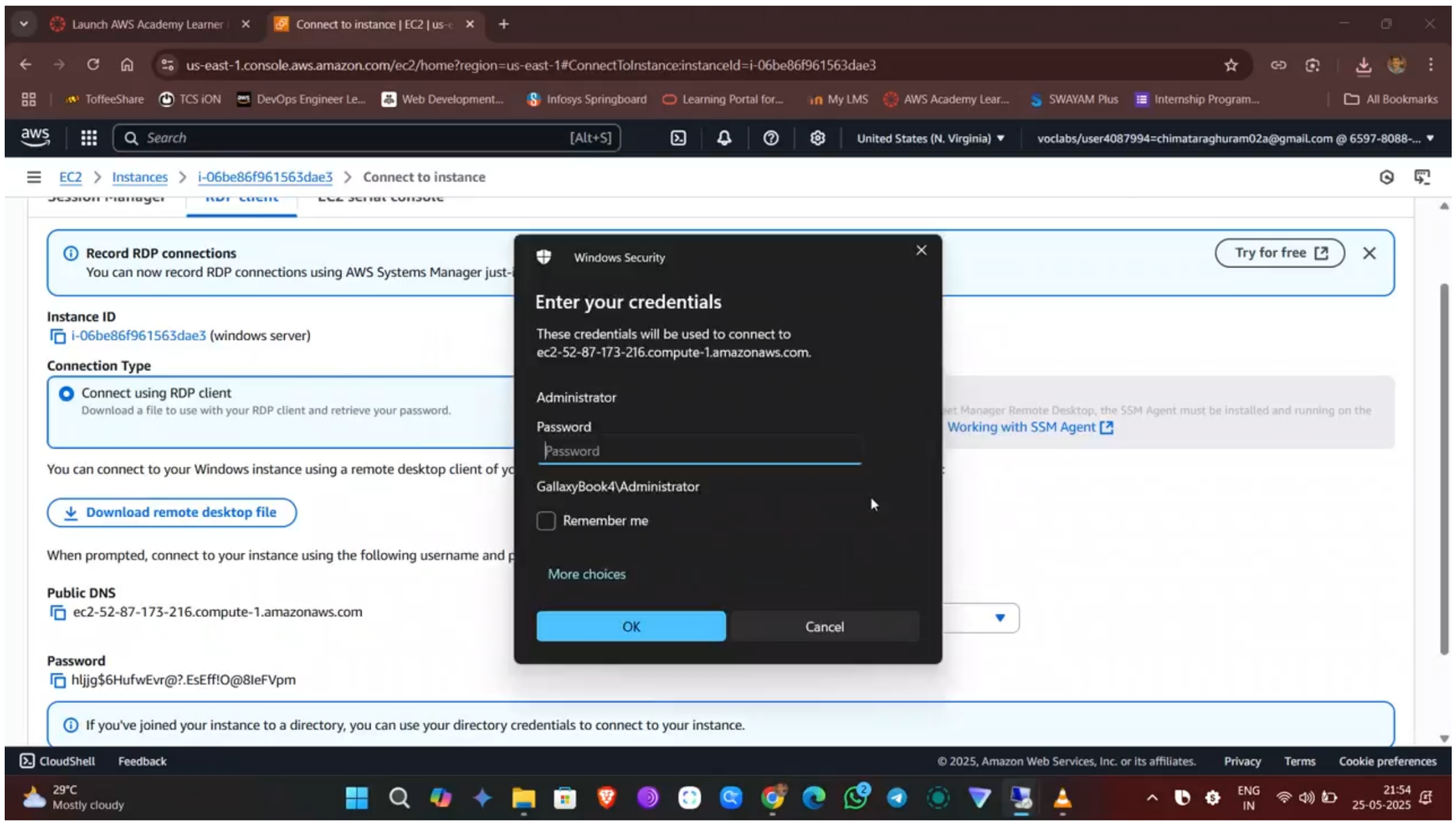
4.2

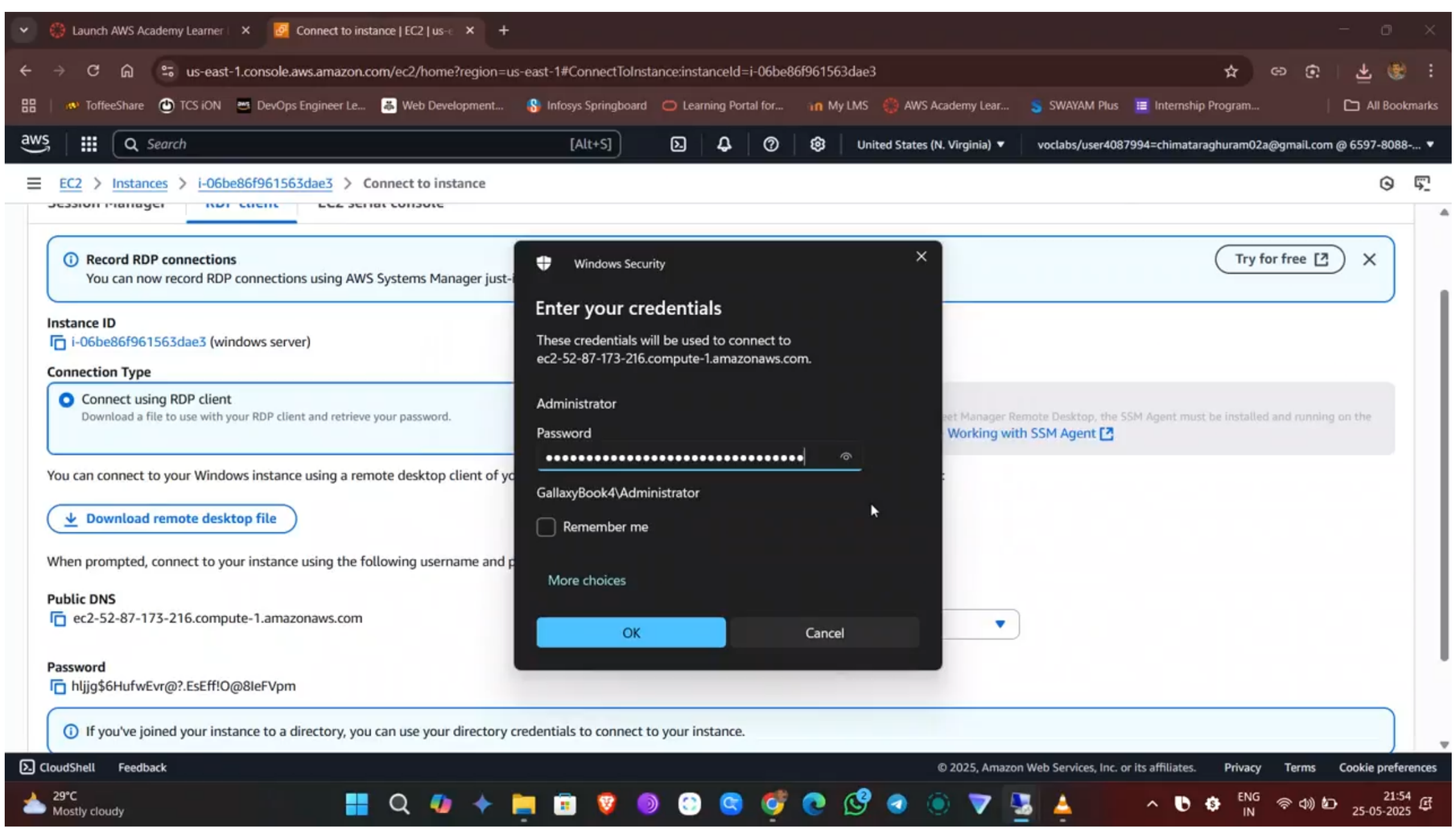
After installing the remote Desktop file some people might get an error just ignore that



4.3

The next step for remote access: Entering the auto-generated password from the AWS console into the Windows Security prompt to establish an RDP session with an EC2 instance



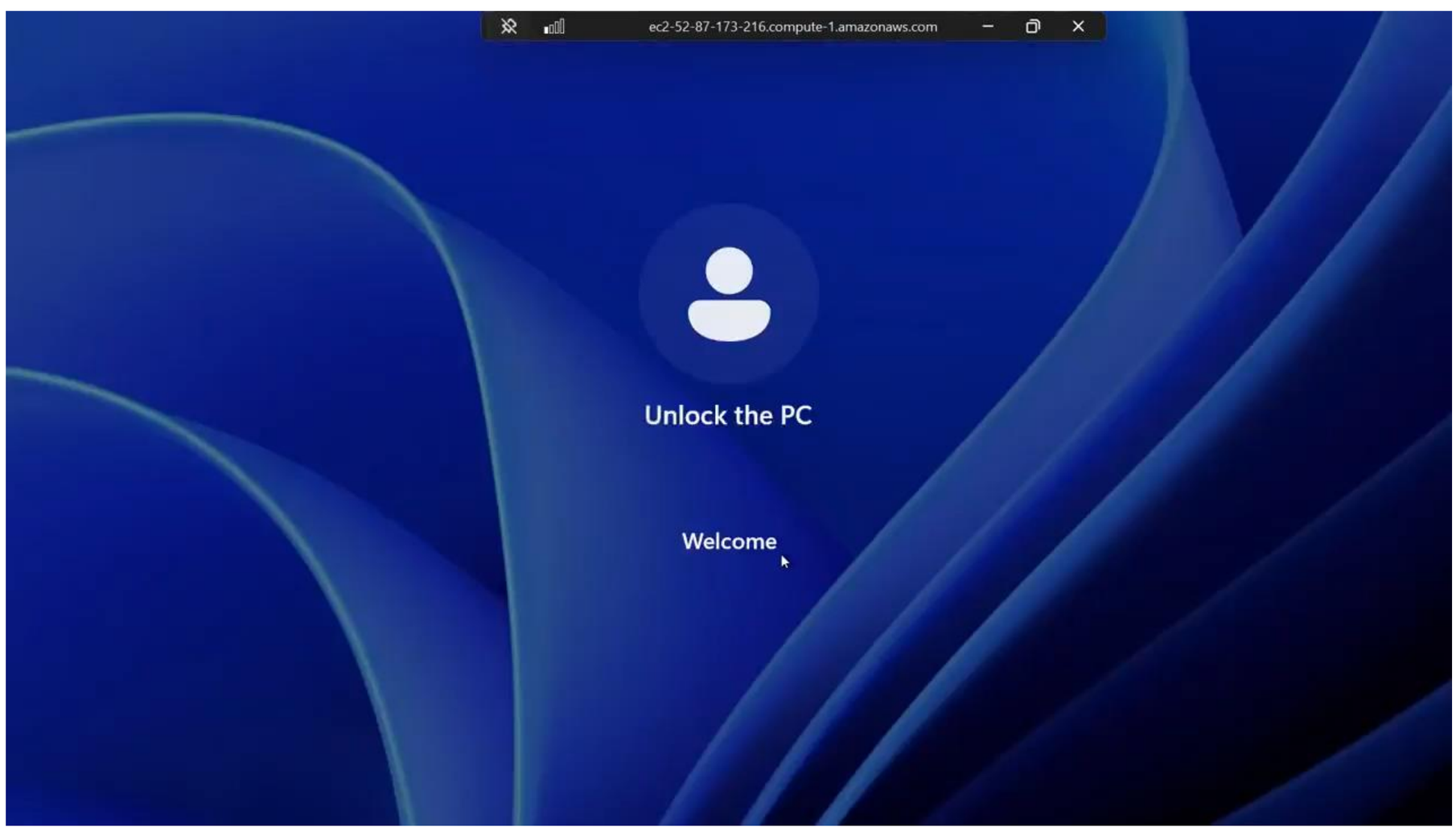


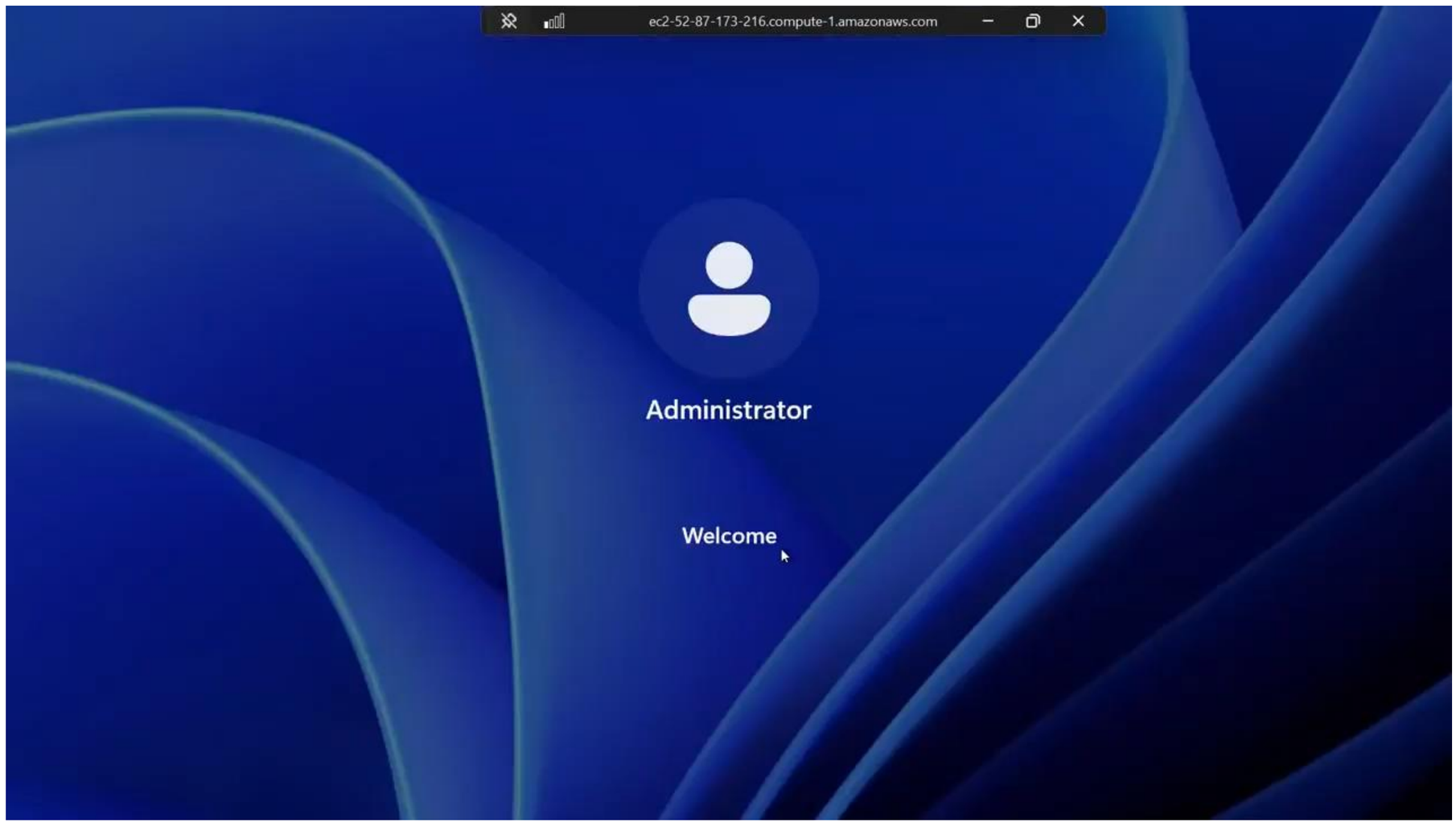
5. Outcome

Successfully created and accessed a virtual Windows desktop hosted on AWS. This setup enables cloud-based desktop access from anywhere with internet connectivity.

5.1

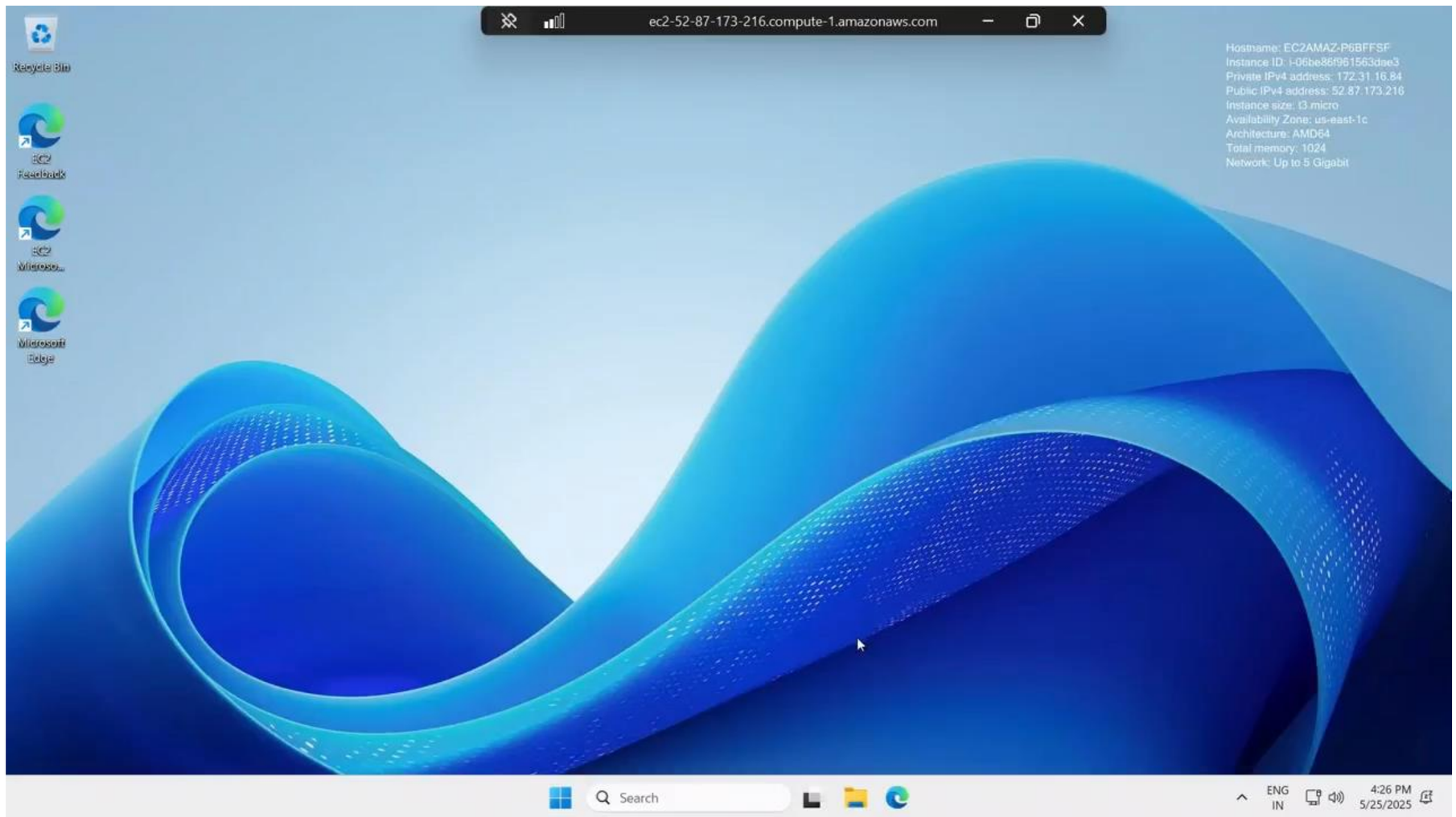
Windows Virtual Desktop login screen for an Administrator on an AWS EC2 instance.





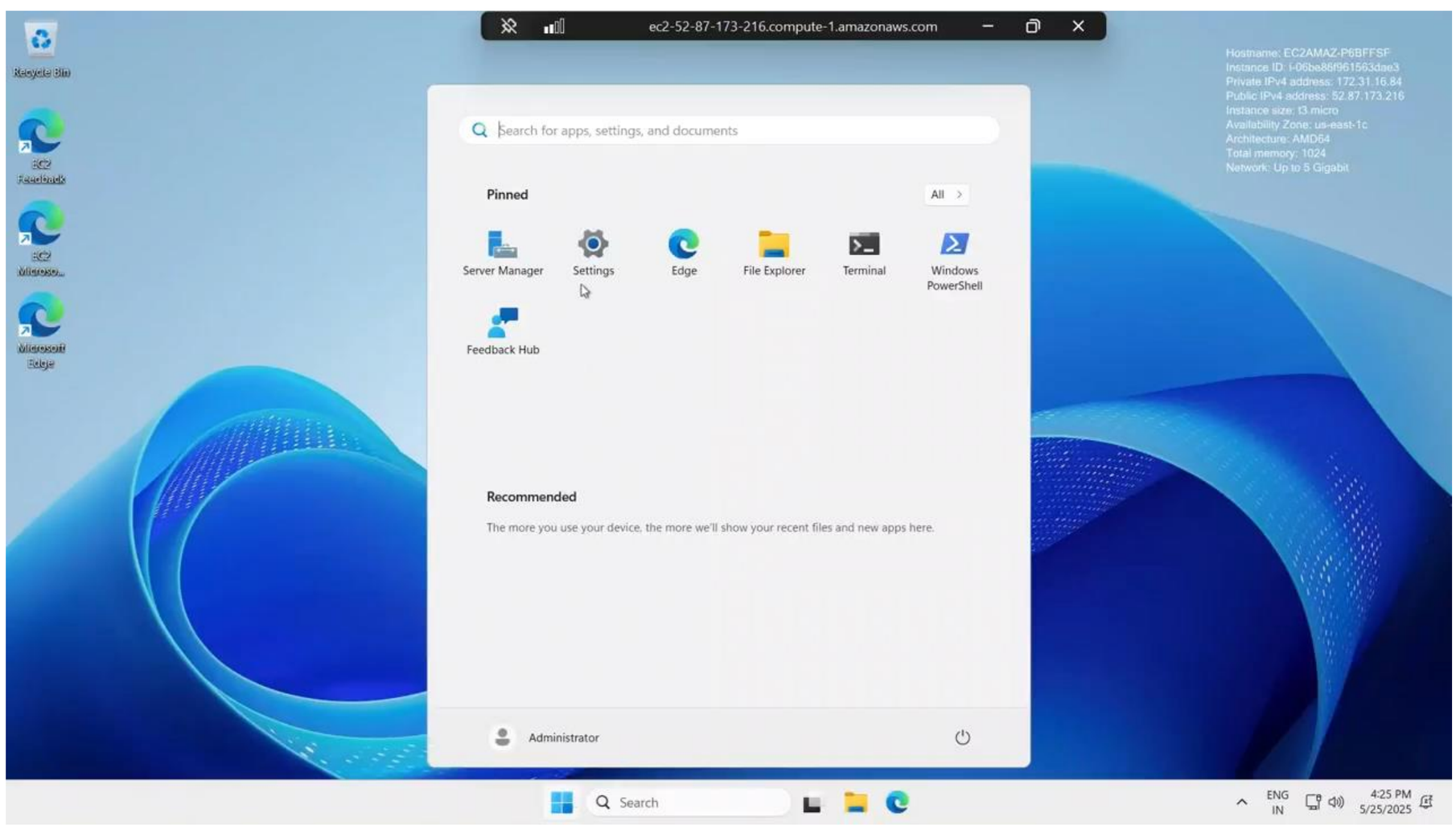
5.3

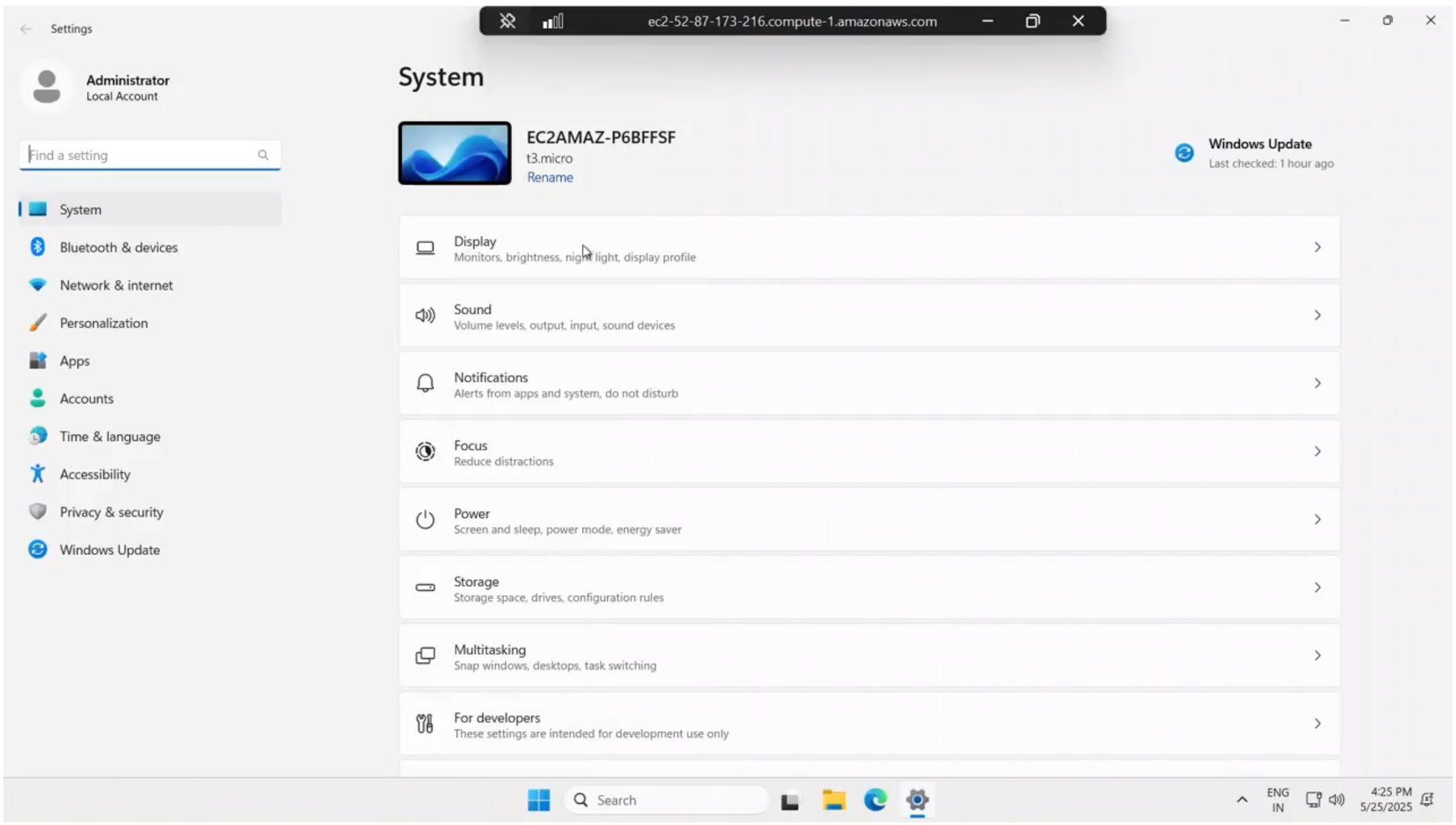
After logging in the wondows Virtual Desktop exploring the home page



5.4

Now exploring some System apps




Settings
ec2-52-87-173-216.compute-1.amazonaws.com
Administrator
Local Account
Find a setting
System
Bluetooth & devices
Network & internet
Personalization
Apps
Accounts
Time & language
Accessibility
Privacy & security
Windows Update
System
EC2AMAZ-P6BFFSF
t3.micro
Rename
Windows Update
Last checked: 1 hour ago
Display
Monitors, brightness, night light, display profile
Sound
Volume levels, output, input, sound devices
Notifications
Alerts from apps and system, do not disturb
Focus
Reduce distractions
Power
Screen and sleep, power mode, energy saver
Storage
Storage space, drives, configuration rules
Multitasking
Snap windows, desktops, task switching
For developers
These settings are intended for development use only
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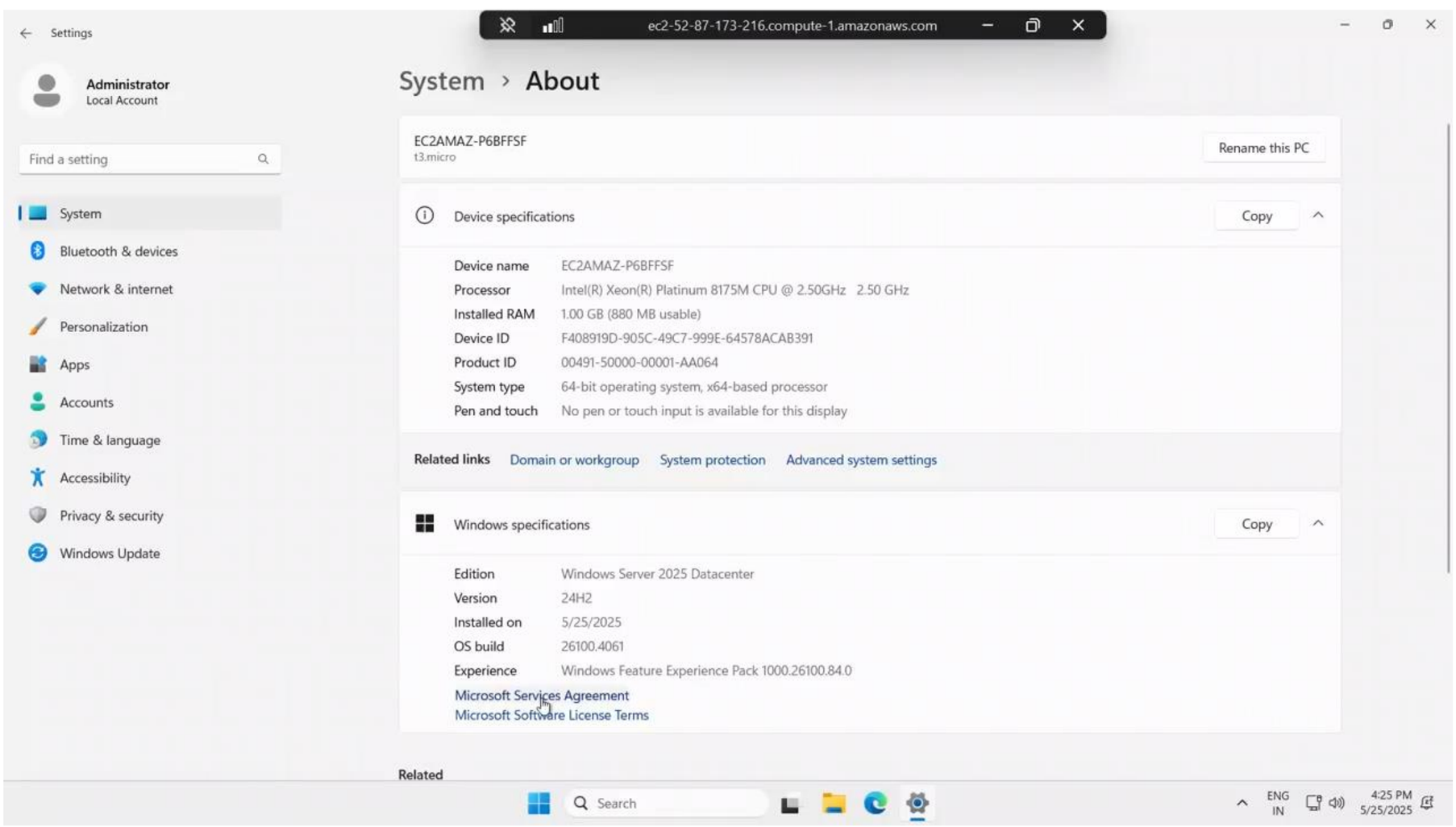
Settings
ec2-52-87-173-216.compute-1.amazonaws.com
Administrator
Local Account
Find a setting
System
Bluetooth & devices
Network & internet
Personalization
Apps
Accounts
Time & language
Accessibility
Privacy & security
Windows Update
System › About
EC2AMAZ-P6BFFSF
t3.micro
Rename this PC
Device specifications
Copy
Device name EC2AMAZ-P6BFFSF
Processor Intel(R) Xeon(R) Platinum 8175M CPU @ 2.50GHz 2.50 GHz
Installed RAM 1.00 GB (880 MB usable)
Device ID F408919D-905C-49C7-999E-64578ACAB391
Product ID 00491-50000-00001-AA064
System type 64-bit operating system, x64-based processor
Pen and touch No pen or touch input is available for this display
Related links Domain or workgroup System protection Advanced system settings
Windows specifications
Copy
Edition Windows Server 2025 Datacenter
Version 24H2
Installed on 5/25/2025
OS build 26100.4061
Experience Windows Feature Experience Pack 1000.26100.84.0
Microsoft Services Agreement
Microsoft Software License Terms
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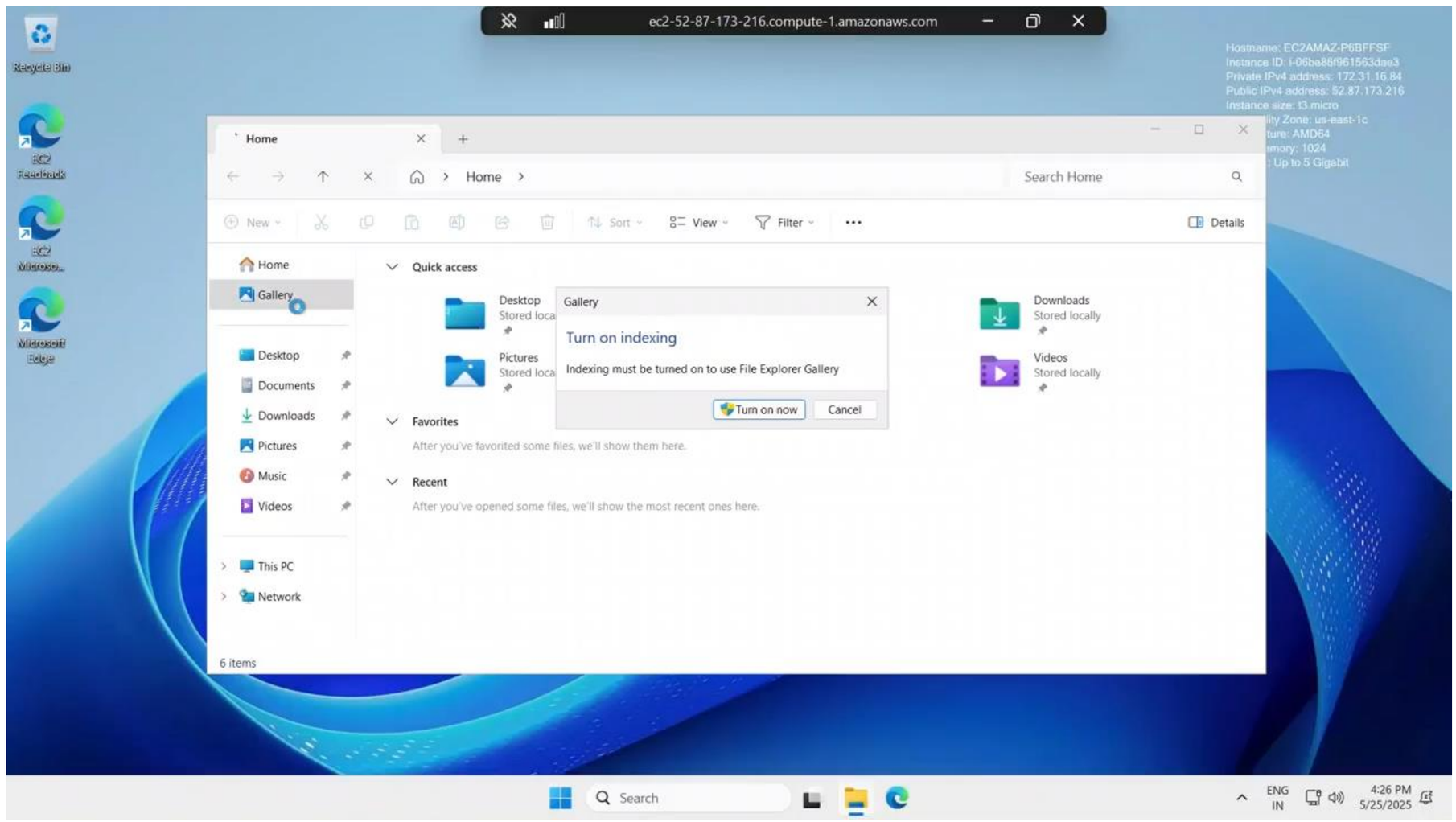
ec2-52-87-173-216.compute-1.amazonaws.com
Hostname: EC2AMAZ-P6BFFSF
Instance ID: i-06be86f961563dae3
Private IPv4 address: 172.31.16.84
Public IPv4 address: 52.87.173.216
Instance size: t3.micro
Recycle Bin
EC2 Feedback
EC2 Microso...
Microsoft Edge
Home
Home
Search Home
New
Sort
View
Filter
Details
Home
Gallery
Desktop
Documents
Downloads
Pictures
Music
Videos
This PC
Network
Quick access
Desktop
Stored loca
Pictures
Stored loca
Downloads
Stored locally
Videos
Stored locally
Gallery
Turn on indexing
Indexing must be turned on to use File Explorer Gallery
Turn on now
Cancel
Favorites
After you've favorited some files, we'll show them here.
Recent
After you've opened some files, we'll show the most recent ones here.
6 items
Search
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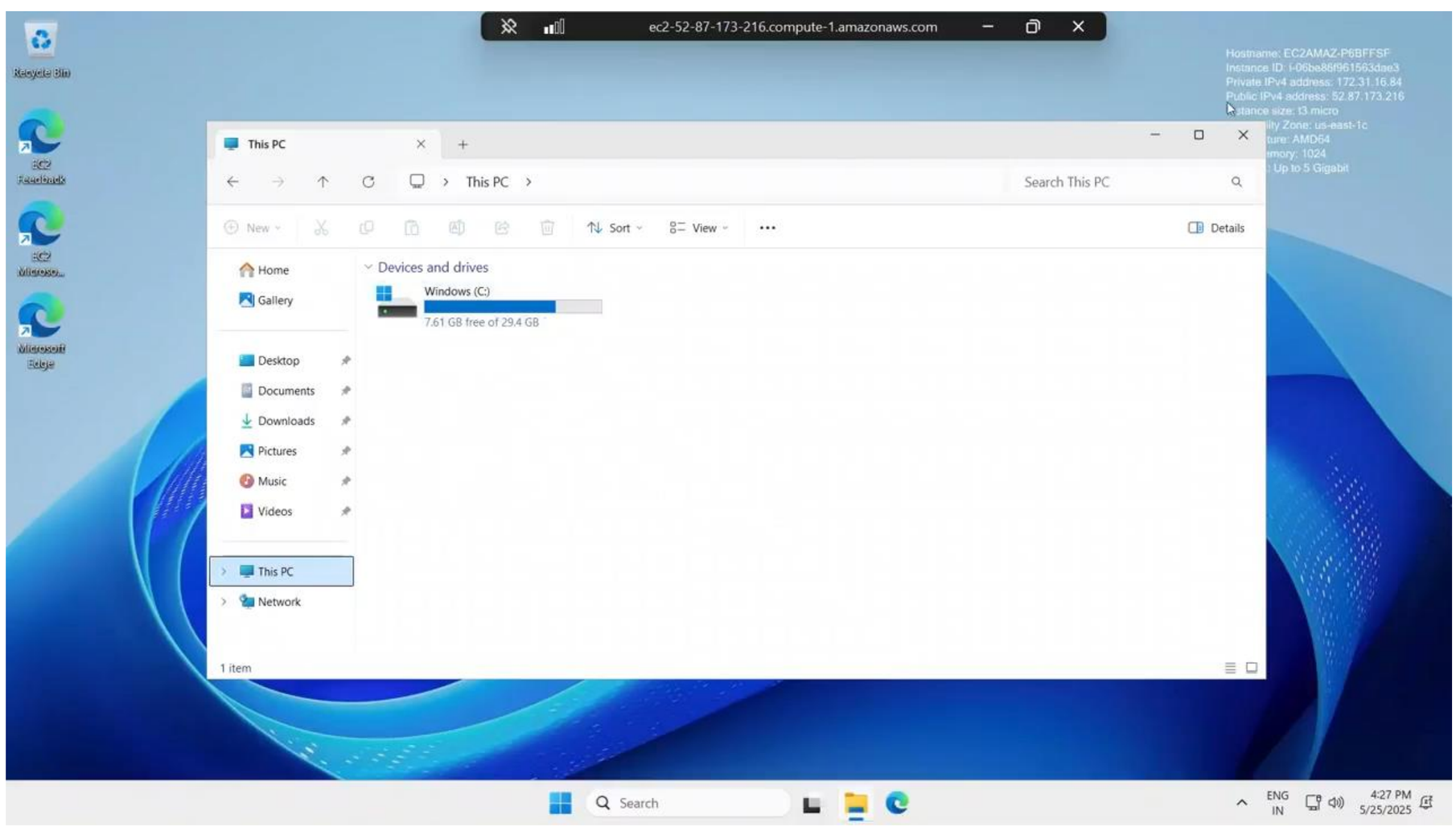
ec2-52-87-173-216.compute-1.amazonaws.com
Hostname: EC2AMAZ-P6BFFSF
Instance ID: i-06be86f961563dae3
Private IPv4 address: 172.31.16.84
Public IPv4 address: 52.87.173.216
Instance size: t3.micro
Recycle Bin
EC2 Feedback
EC2 Microso...
Microsoft Edge
This PC
This PC
Search This PC
New
Sort
View
Details
Home
Gallery
Desktop
Documents
Downloads
Pictures
Music
Videos
This PC
Network
Devices and drives
Windows (C:)
7.61 GB free of 29.4 GB
1 item
Search
ENG
IN
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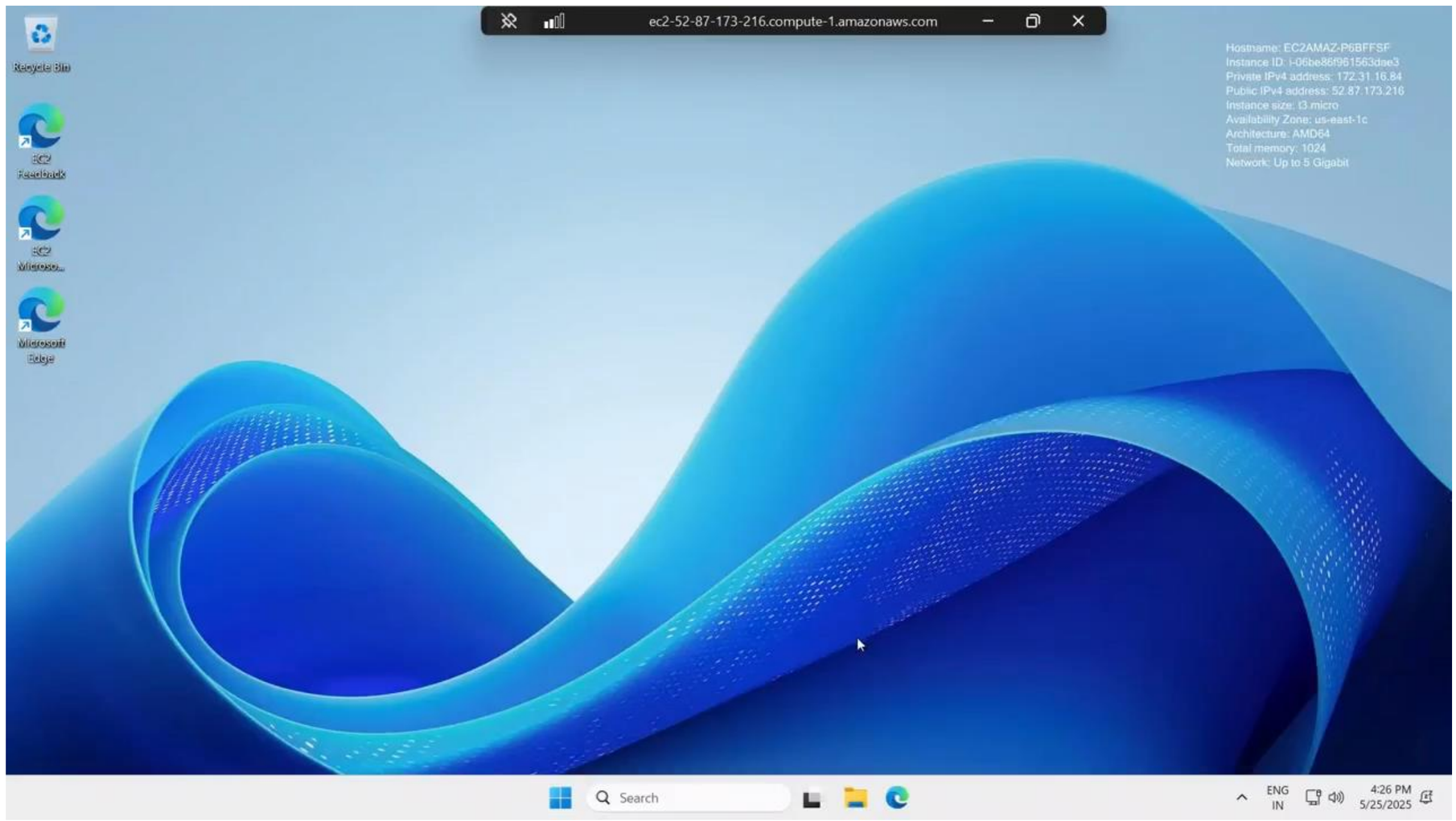

6. Key Learnings

This project provided valuable insights into:

- Basic cloud infrastructure and deployment
- Virtual networking and security group configuration
- Remote desktop setup and troubleshooting

These skills are foundational for roles in full-stack development, DevOps, and system administration.

7. Conclusion

Overall, this mini project helped strengthen foundational cloud computing skills and gave practical exposure to deploying and managing remote desktop environments on AWS.